

Figure 3 Gate

Two Framings of the Single-Deletion Multimodal Figure, as Mockups

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Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 The decision ✗

Figure 3 covers single-gene deletions read out in three modalities: the transcriptome (Kemmeren 2014, 1,484 deletions), the proteome (Messner 2023, 4,476) and morphology (Ohya 2005, CalMorph, 4,718). Two framings are open and they send the next compute to different places. Framing A is a conditioning figure: one modality is measured and another is predicted from it, which is an imputation claim and needs the conditioning measurement for every strain the model is applied to (fig. 1). Framing B is a genotype-only figure: what the deletion alone predicts against each replicate ceiling and against linear and nearest-neighbor baselines (fig. 2). The choice is A, B, or neither; figs. 3 and 4 hold the supporting panels, none of which repeats a main panel, and table 1 lists what each panel lacks. The joint-training round v19 is a null on both registered tests. The conditioned round v20 is in flight at 2 of 12 split seeds and about 160 of 1,200 epochs, so framing A currently rests on the ridge triangle and on a partial run. On file, the best nearest-neighbor baseline is within 0.01 of the best model score on expression and above it on the proteome and on morphology, each a maximum over keys selected on validation (fig. 1b). SRC: `experiments/019-simb-multimodal/results/joint_checkpoint_readout.json`

Status marks. Every panel letter carries the status glyph used for section headings here: ✓ ready, the panel can be drawn now from committed results; ■ partial, or waiting on an author decision; ✗ not done, a placeholder. The marks are mockup annotations on their own draw.io layer and are switched off with `FIG3_STATUS_MARKS=0`. A *split seed* is a data partition: each seed is a different train, validation and test split, run with one initialization seed.

2 The mockups ✗

Result files are under `experiments/019-simb-multimodal/results/` unless a path is given. Each caption gives, per panel, what is plotted, the statistic and its scoring rule, the sample size and the status; the table under each figure gives the files and keys.

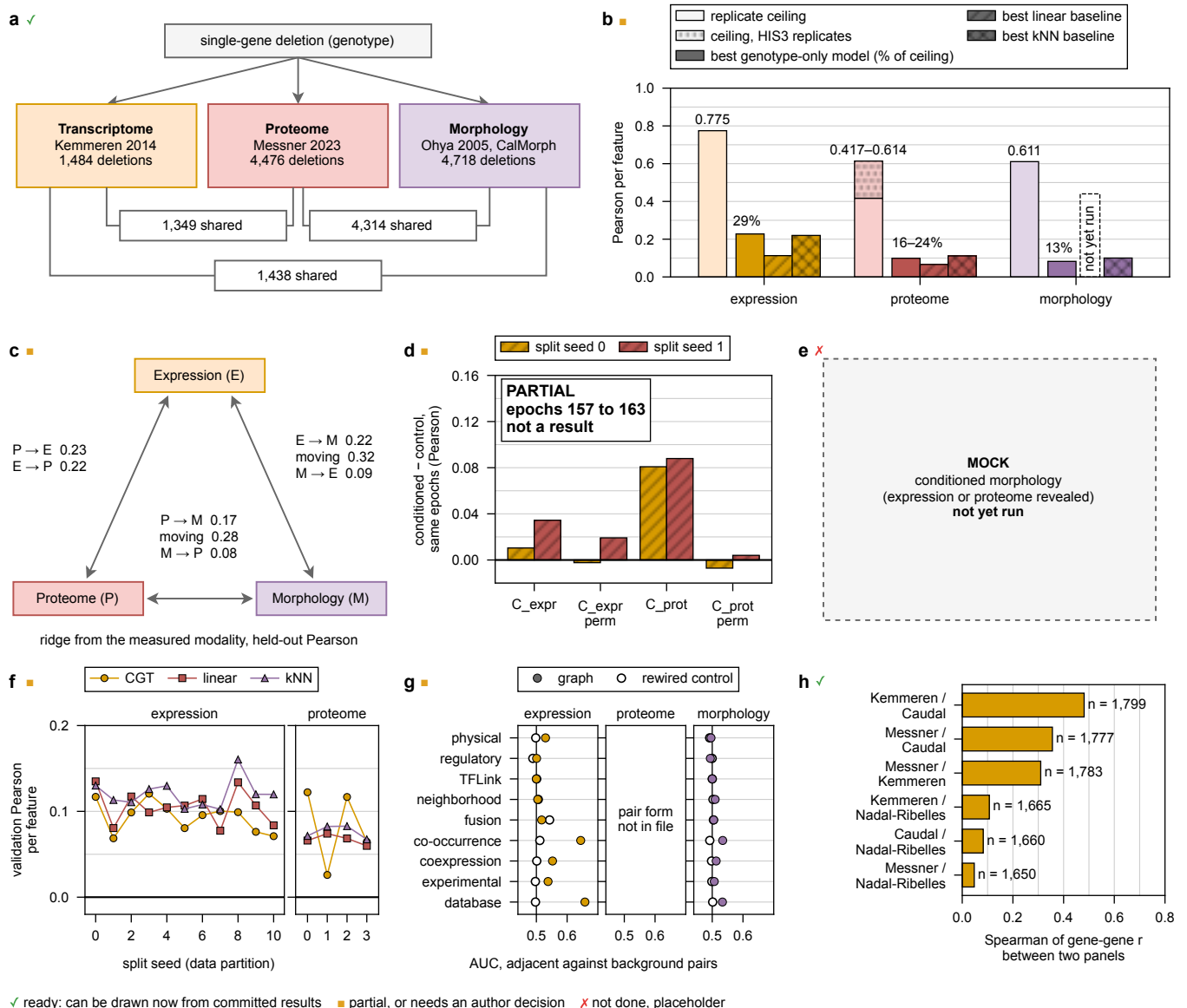
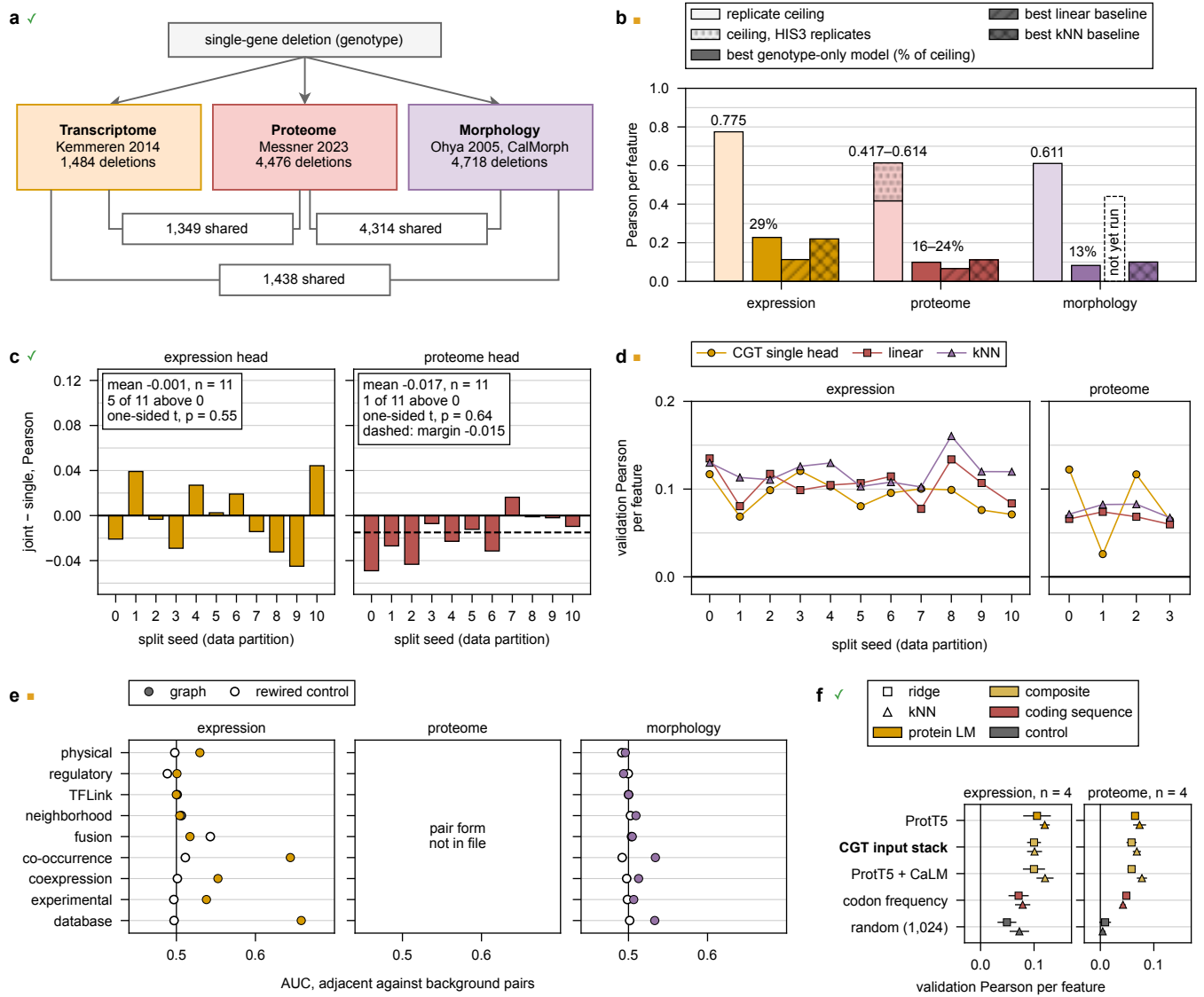


Figure 1: Ridge from one measured modality reaches a held-out Pearson of 0.17 to 0.23 on all features of another, and the conditioned model (d) has trained 160 of 1,200 epochs. **a**, The three single-deletion panels and their pairwise overlaps. Counts of distinct deletion strains; no statistic. Ready. **b**, Replicate ceiling (pale; dotted, the HIS3-replicate proteome estimator), best genotype-only model (solid; percent of ceiling), best linear baseline (diagonal hatch) and best nearest-neighbor baseline (cross hatch), as Pearson per feature across held-out strains. Model: expression and morphology are the rolling maximum of one best run over 1,353 and 183 runs, the proteome is the mean over 4 split seeds. Baselines: mean validation score over 4 split seeds of the best key (morphology, 1 split seed); no linear baseline exists for morphology. Every non-ceiling bar is a maximum over runs or keys selected on the score it receives, so it is biased upward, and the bars sit on different splits. Partial. **c**, Out-of-fold ridge between measured modalities on the deletions both panels measured (1,349, 4,314 and 1,438 strains; 5 folds): median held-out Pearson per feature; *moving* is the 116 morphology features with replicate reliability of at least 0.5. The proteome and expression side is a mean per feature from a review file. Partial. **d**, Round v20, conditioned arm minus unconditioned control: difference of the mean validation Pearson over the latest 20 epochs of each conditioned run and the same epochs of its control; 2 of 12 split seeds, 8 runs, epochs 157 to 163 of 1,200. Not a result. Partial. **e**, Conditioned morphology. Not run. Not done. **f**, Linear and nearest-neighbor baselines and the v19 single-head model per split seed (expression 11, proteome 4). Baselines: one validation Pearson per split seed, best embedding selected on that score (biased upward). Model: mean validation Pearson over the registered epoch window (unselected). A split seed is one train, validation and test partition with one initialization seed; baselines and model share the seed number and not the store, so their validation strains differ. Partial. **g**, Per graph, the AUC that two deletions adjacent on the graph give more similar phenotypes than a background pair (filled), against a degree-preserving rewired graph (open); 60,000 pairs per point, 1,482 expression and 4,718 morphology strains; no proteome strand is on file. Partial. **h**, Spearman between two panels' gene-by-gene correlation matrices, with the number of shared genes; Messner is the proteome, the others mRNA. One value per pair, no selection. Ready.

Sources for fig. 1

a	fig3_overlap_census.json (kemmeren_unique, ohya_unique); fig3_proteome_build_census.json; n_shared_strains of the two covariation files
b	expression_ceiling_replicate.json, proteome_ceiling_replicate.json, morphology_noise_ceiling.json, round_leaderboards.csv; baselines_embedding_study.json (ridge 0.113, 0.066); graph_retrieval_baseline.json (nearest neighbor 0.220, 0.112, STRING experimental graph key); knn_embedding_probe.json (0.100)
c	proteome_morphology_covariation.json (and its context block), expression_morphology_covariation.json
d	joint_checkpoint_readout.json, v20_partial.runs
f	expression_baselines_split/, baselines_split_fig3_proteome/; joint_checkpoint_readout.json, v19.runs
g	graph_prior_probe.json, pair_form, one step
h	experiments/028-knockout-expression/results/proteome_expression_covariation.json, gene_covariation

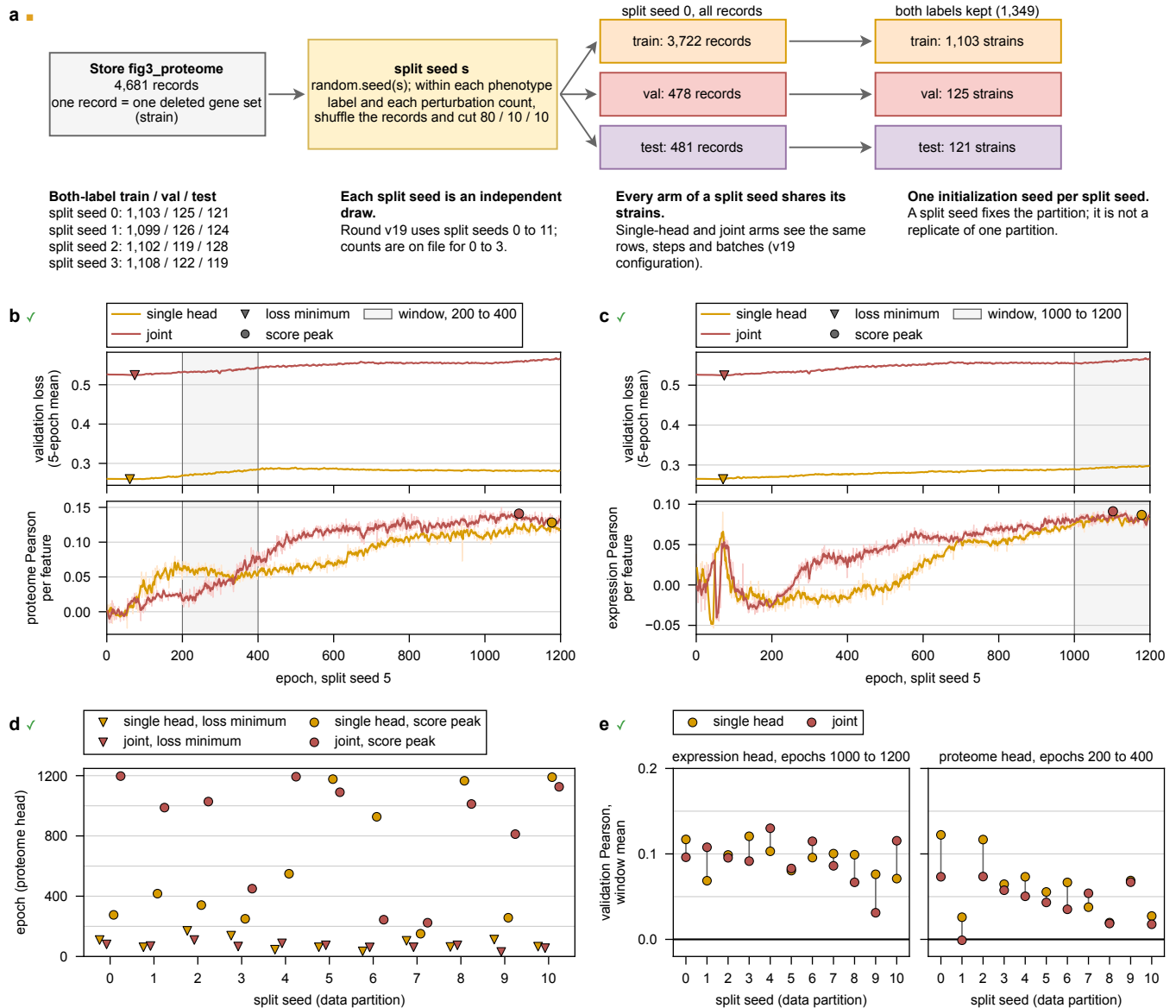


✓ ready: can be drawn now from committed results ■ partial, or needs an author decision ✗ not done, placeholder

Figure 2: From the deletion alone the model reaches 13 to 29% of the replicate ceiling, and the best nearest-neighbor baseline is within 0.01 of it on the proteome and above it on the morphology. **a**, **b**, As fig. 1a and b. Ready; partial. **c**, Round v19, joint arm minus single-head arm per split seed: difference of the mean validation Pearson per feature over the registered window (expression epochs 1,000 to 1,200, proteome 200 to 400), an unselected statistic. 11 of 12 split seeds, each a different train, validation and test partition with one initialization seed; 1,099 to 1,108 training and 119 to 126 validation strains on the four seeds on file. Expression: mean -0.001 , 5 of 11 above 0, one-sided t $p = 0.55$. Proteome: mean -0.017 , 1 of 11, $p = 0.64$ against the margin -0.015 (dashed). Neither registered test rejects. Ready; split seed 11 was unfinished. **d**, As fig. 1f at half width, same caveat. Partial. **e**, As fig. 1g at wide width. Partial. **f**, How much the gene representation matters to the two baselines: validation Pearson per feature, mean and s.d. over 4 split seeds, of the ridge baseline (squares) and the nearest-neighbor baseline (triangles), with rank and neighbor count selected on validation inside each seed. Five of 28 representations by rule, each the top of its class on the mean of the four validation means: best single language model (ProtT5), the model's input composite (bold), best other composite, codon frequency, best random control. The bold row is the study's **model_stack**, the concatenation **fudt_upstream**, **calm**, **prot_T5_all**, **fudt_downstream** that the v19 configuration inherits as **node_embeddings**. Ready.

Sources for fig. 2

- c `joint_checkpoint_readout.json`, `v19.tests` (`H1a_expression_superiority`, `H1b_proteome_noninferiority`); strain counts from `split_gene_overlap_audit.json`
 - f `baselines_embedding_study.json`; the composite is defined in `experiments/019-simb-multimodal/scripts/expression_baselines_split.py` and matched to `conf/cgt_expr_v13_split.yaml`
-



✓ ready: can be drawn now from committed results ■ partial, or needs an author decision ✗ not done, placeholder

Figure 3: In round v19 the validation loss reaches its minimum before epoch 200 on every run, and the proteome score peaks later on every run. **a**, How a split seed becomes the strains of a run, drawn from the code. The store's 4,681 records (one record is one deleted gene set) are shuffled within each phenotype label and each perturbation count and cut 80, 10, 10 under the split seed; each side then keeps the records carrying both labels, 1,349 in all, so the both-label fractions are near and not exactly 80, 10, 10. Counts for split seeds 0 to 3. No statistic. Partial: seeds 4 to 11 are not on file. **b**, Proteome head, split seed 5, one run per arm: the 5-epoch centered mean of the validation loss (top) and the validation Pearson per feature (bottom; pale raw, dark its 5-epoch mean) against epoch. Triangles, loss minimum; circles, maximum of the 5-epoch mean, an upward-biased order statistic; shaded, the registered window. The joint loss sums two heads, so the two loss levels are not comparable. Split seed 5 is chosen by rule: its joint-minus-single proteome difference (-0.012) is the median over the 11 complete split seeds. About 1,100 training and 125 validation strains. Ready. **c**, The same for the expression head of that split seed, window epochs 1,000 to 1,200. Ready. **d**, Proteome head, all 11 complete split seeds, 22 runs: epoch of the loss minimum (triangles) and of the score peak (circles). Ready. **e**, The window scores behind fig. 2c: mean validation Pearson over the registered window, single-head and joint arm, 11 split seeds, 33 runs. Unselected. Ready.

Sources for fig. 3

a	torchcell/datamodules/cell.py	(<code>_compute_and_save_index</code>);	experiments/019-simb-multimodal/scripts/train_cgt_multitask.py	(<code>require_modalities</code>); counts from <code>split_gene_overlap_audit.json</code>
b, c	joint_checkpoint_curves.csv	(41,129 rows: run, arm, split, epoch, loss, its 5-epoch mean, each head's Pearson), written by experiments/019-simb-multimodal/scripts/joint_checkpoint_readout.py; marker epochs from joint_checkpoint_readout.json		
d, e	joint_checkpoint_readout.json, v19.runs			

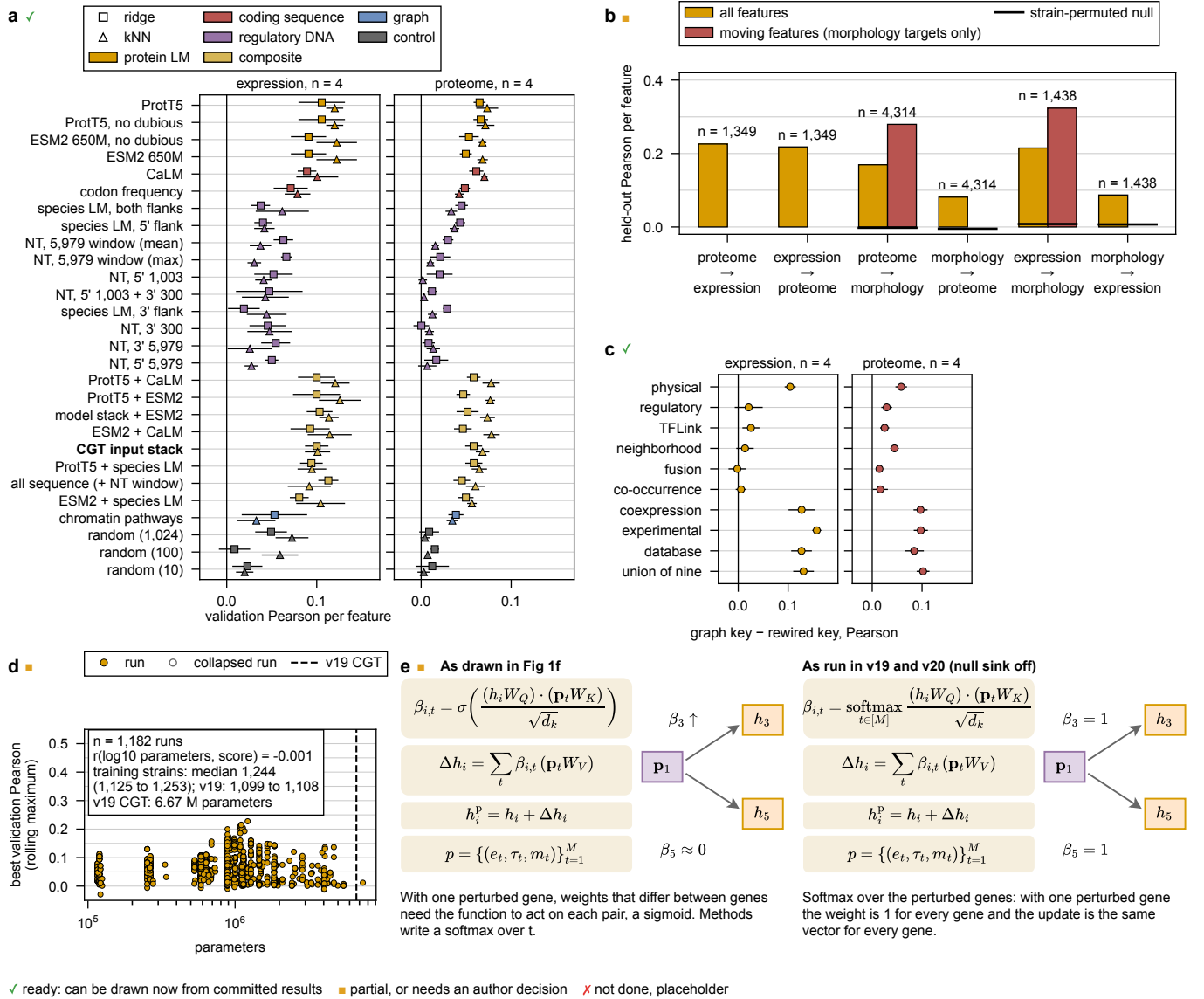


Figure 4: The model's input composite scores below ProtT5 alone in both baselines on both modalities, and the best score of a run does not track its parameter count. **a**, All 28 gene representations of the baseline study, by family: validation Pearson per feature, mean and s.d. over 4 split seeds, ridge (squares) and nearest neighbor (triangles), expression (1,244 to 1,253 training strains) and proteome (3,581); rank and neighbor count selected on validation inside each seed. Bold, the model's input composite. Ready. **b**, The six directed pairs of fig. 1c as bars with the strain-permuted null (black line, on file for four pairs) and the shared-strain count. Partial. **c**, The graph as a retrieval key: validation Pearson of a neighbor average keyed on each graph's adjacency row minus the same with the rewired graph, mean and s.d. over 4 split seeds, expression and proteome. Neighbor count selected on validation. Ready. **d**, Parameter count against the rolling maximum of the validation Pearson, one point per expression-strand run (1,182 runs with both values; open, collapsed runs). The rolling maximum is biased upward and the runs differ in epochs run. The Pearson between $\log_{10}$ parameters and score is -0.001 . Dashed, the v19 model at 6.67 million parameters for about 1,100 training strains, near 6,000 parameters per strain; its score is a window mean and is not placed on this axis. Partial. **e**, The perturbation operator in the notation of manuscript Fig. 1f; no data. Left, as drawn there. Right, as the code computes it and as the Methods write it: a softmax over the perturbed genes, which with one perturbed gene gives weight 1 to every gene. Partial, an author decision.

Sources for fig. 4

- a `baselines_embedding_study.json`
 - b as fig. 1c, plus triangle of `joint_checkpoint_readout.json` for one strain count
 - c `graph_retrieval_baseline.json`, rounds v13 and v14, `summary.paired`
 - d `round_leaderboards.csv` (`total_param_count`, `primary_roll_max`); `total_param_count` of v19.runs in `joint_checkpoint_readout.json`; `split_gene_overlap_audit.json`
 - e `torchcell/models/equivariant_cell_graph_transformer.py` (`EquivariantPerturbationTransform`); `paper/nature-biotech/sections/methods.tex`
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3 What each panel still needs ✗

Table 1: What each panel still needs before it is real. ✓ ready: every drawn value is in a committed results file and the panel can be drawn now. ■ partial: a run is unfinished, scoring rules or splits are mixed, or an author decision is pending. ✗ not done: nothing behind it has been run.

panel		what remains
A and B, a	✓	The expression and morphology overlap reads 1,438 in <code>expression_morphology_covariation.json</code> and 1,440 in <code>fig3_overlap_census.json</code> ; reconcile.
A and B, b	■	No linear baseline for morphology is on file. The bars mix scoring rules and splits. Rescore model and baselines under one rule on one set of split seeds, and choose one proteome ceiling estimator.
A, c; SI-2, b	■	The proteome and expression side is a mean per feature from a review file, with no permuted null and no moving subset on file; recompute it with the estimator of the other two sides, on a common strain set.
A, d	■	2 of 12 split seeds at epochs 157 to 163 of 1,200. Needs the full budget, all split seeds, the registered window, and the ridge of panel c as the reference to beat.
A, e	✗	No store joins the proteome and morphology, and the observed-label channel takes gene-level values, so morphology can be a target and not yet an input.
A, f; B, d	■	Baselines and model share a seed number and not a store, and the baselines are single selected validation scores while the model is a window mean. Rerun the baselines on the both-label store. The proteome baselines cover split seeds 0 to 3 only.
A, g; B, e	■	The pair-form probe has no proteome strand; add it to <code>graph_prior_probe.py</code> .
A, h	✓	Sameith is in <code>gene_covariation_all.json</code> and not drawn.
B, c; SI-1, e	✓	Split seed 11 was unfinished at readout; redraw when it is read. The result is a null.
B, f; SI-2, a	✓	Four split seeds; the baselines ran on the full expression and proteome stores.
SI-1, a	■	Both-label counts are on file for split seeds 0 to 3 of 12. That every arm sees the same batches is stated by the v19 configuration and was not re-verified here.
SI-1, b, c	✓	One split seed by rule; the other ten are in <code>joint_checkpoint_curves.csv</code> .
SI-1, d	✓	The score peak is a rolling maximum, an upward-biased order statistic.
SI-2, c	✓	Four split seeds per strand.
SI-2, d	■	The expression document's "145 runs, -0.129" is produced by no committed script and no subset of <code>round_leaderboards.csv</code> tried here reproduces it; the panel draws all 1,182 scored expression runs, $r = -0.001$. Scores are rolling maxima over unequal budgets. A v19 point needs a v19 score under the same rule.
SI-2, e	■	Author decision: Fig. 1f draws gene-dependent weights with one perturbed gene, and the Methods and the code compute a softmax, which gives weight 1. Either the figure or the operator changes. The null sink, which restores a sigmoid weight, has one trained comparison stopped at 186 to 189 epochs.